

Huili Zhang

Research Associate
Department of Geoscience,
University of Wisconsin-Madison, USA

Tel: (+86) 18851087566
E-Mail: 2970143759@qq.com;
h Zhang2465@wisc.edu



Education

-
- | | |
|---|-------------------------|
| Ph.D., Geology, Nanjing University | 09/2020–09/2024 |
| <ul style="list-style-type: none">• The properties of the mantle source of Cenozoic basaltic rocks• Volatile (CO₂, F, and Cl) cycle between the deep mantle and crust | |
| <p style="text-align: right;">Supervisor: Gang Zeng, Jin-Hai Yu
Rocking Mantle Group (http://rockingmantle.com/)</p> | |
| M.S., Petrology, Nanjing University | 09/2018– 06/2020 |
| <ul style="list-style-type: none">• The properties of the mantle source of Cenozoic basaltic rocks | |
| <p style="text-align: right;">Supervisor: Gang Zeng
Rocking Mantle Group (http://rockingmantle.com/)</p> | |
| B.S., China University of Petroleum | 09/2014– 07/2018 |

Project/Grants

Improvement of a method for in-situ F-Cl-Nb-P analysis in melt inclusions (10,000 ¥; 2022.3-2023.3, PI).

Experience and skills

-
- **Skilled in EPMA:**
 - Developed the method of trace F-Cl-Nb-P analysis in melt inclusions (MI) by EPMA;
 - Skilled in the major- and minor- elemental analysis of minerals and glasses;
 - Completed the EPMA training course (JEOL, Japan) and got their certification.
 - **Familiar with a wide range of computational and analytical tools:**
 - Petrolog3**, **MELTS**, Office, Illustrator, SigmaPlot, Python, C++, CorelDRAW, origin;
 - SEM** (gain high-resolution images);
 - LA-ICP-MS** (trace element in minerals and melt inclusions (MI), Pb isotope in MI);
 - Rama spectrometry** (identify the types of minerals and volatiles in melt inclusions);
 - Conventional box furnace** (synthesize reference materials).
 - **Experience as the assistant for the “Petrology” course in Nanjing University and the assistant in State Key Laboratory of EPMA.**

Publications

Zhang, H.-L., Zeng, G., Zhang, C., Zhang, W.-L., Chen, L.-H., and Yu, J.-H., **2022**, Electron Probe Microanalysis measurement of F-Cl-Nb-P for Geological Glasses: **Geostandards and Geoanalytical Research**, v. 46, no. 4, p. 851-864.

Zhang, H.-L., Zeng, G., Liu, J.-Q., Chen, L.-H., Yu, J.-H., Wu, B., Wang, X.-J., Xu, X.-S., and Liu, X.-W., **2023**, Carbonated eclogitic component beneath eastern China revealed by olivine phenocrysts in nephelinites: **Chemical Geology**, v. 640, p. 121744.

Zhang, H.-L., Zeng, G., Chen, L.-H., Liu, J.-Q., Yu, J.-H., Xu, X.-S., 2024. Evaluation of CO₂ concentration in a carbonated eclogite mantle source: A new attempt based on the compositions of olivine phenocrysts. **Lithos**, v. 482-483, p. 107669.

Zeng, G., **Zhang, H.L.**, Liu, J.Q., Chen, L.H. Deep cycling of fluorine and chlorine traced by melt inclusions trapped in olivine phenocrysts from the Hainan Island basalts. **Bulletin of Mineralogy, Petrology and Geochemistry**, 2023,42(06):1248-1259+1210.
<https://doi.org/10.19658/j.issn.1007-2802.2023.42.103>.

Shi, J.H., Zeng, G., Chen, L.H., Wang, X.J., Liu, J.Q., Xie, L.W., Yang, Y.H., **Zhang, H.L.**, **2023**. Lithology of EM1 reservoir revealed by Fe isotopes of continental potassic basalts. **Journal of Geophysical Research: Solid Earth**, 128(1): e2022JB025133.
<https://doi.org/10.1029/2022JB025133>.

Liu, J.Q., Chen, L.H., Wang, X.J., **Zhang, H.L.**, Zeng, G., Saskia, E., Zhang, L., Ren, Z.Y. **2022**. Olivine and melt inclusion chemical constraints on the nature and origin of the common mantle component beneath eastern Asia. **Contributions to Mineralogy and Petrology**. 177:116(2022)
<https://doi.org/10.1007/s00410-022-01981-y>.

Liu J. Q., Erdmann, S., Chen L. H., **Zhang H.L.**, Wu B., Zeng G., Wang X. J., Lei Z. L., & Yu X. (**2021**). Petrological evidence for magma recharge and mixing beneath the Ma'anshan monogenetic volcano of Xiaogulihe in Northeast China. **Lithos**. 382-383.
<https://doi.org/10.1016/j.lithos.2020.105928>.

Presentations

Extremely carbon-rich mantle beneath eastern China revealed by olivine phenocrysts within nephelinite; Oral presentation on “*Goldschmidt 2023*”; Lyon, France; July 2023.

Electron Probe Microanalysis measurement of F-Cl-Nb-P for Geological Glasses; Oral presentation on “The 8th Youth Science Forum”; Wuhan, China; May 2023.

Electron Probe Microanalysis measurement of F-Cl-Nb-P for Geological Glasses; Oral presentation on “The Third National Geoscience Graduate Forum in 2022”; Online, December 2022.

Tracing the CO₂-rich mantle using the olivine phenocrysts and their melt inclusions; Oral presentation on “The 7th Youth Science Forum”; Guiyang, China; July 2021.

Honors, Awards (reverse chronological order)

<i>Excellent graduate in 2024</i>	2024
<i>Excellent graduate student in 2023</i>	2023
<i>Industrial bank scholarship in 2023</i>	2023
<i>Second Prize of the Meritocracy Scholarship in 2022</i>	2022
<i>Excellent report award in “The Third National Geoscience Graduate Forum in 2022”</i>	2022
<i>First-class scholarship for graduate study</i>	2019
<i>The first prize of excellent scholarship for learning</i>	2016-2017
<i>National Encouragement Scholarship</i>	2015-2016
<i>The third prize of excellent scholarship for learning</i>	2014-2015
